

Deva Anand

(413) 315-1715 | devaanand@umass.edu | devaanand.com | linkedin.com/in/devaaa | github.com/Deva-1903

Education

University of Massachusetts Amherst

Master of Science in Computer Science

September 2025 – May 2027 (Expected)

Amherst, MA

Coursework: Advanced Machine Learning, Advanced NLP, Optimization in CS, Systems for Data Science, Performance Engineering

Vels Institute of Science, Technology and Advanced Studies

Bachelor of Engineering in Computer Science and Engineering

August 2019 – May 2023

Chennai, India

Skills

Languages

Python, JavaScript, TypeScript, C++, Java, SQL, Bash

Machine Learning

PyTorch, JAX, HuggingFace Transformers, NumPy, pandas, scikit-learn, transfer learning, normalizing flows, diffusion, autodiff

Data & Pipelines

Apache Spark / PySpark, ETL & ingestion, schema-aware reads, partitioning, data-quality filtering, Spark UI, PostgreSQL, MongoDB, SQLite

Retrieval & LLM

RAG over vector stores, dense retrieval, sentence-transformers, RRF, cross-encoder reranking, MMR, LLM agents, OpenAI / Anthropic / Groq APIs, Ollama

Backend & Cloud

FastAPI, Node.js, REST APIs, Docker, AWS, GCP, Azure, Git, GitHub Actions, CI/CD

Experience

Wysa

Backend Engineer

July 2023 – July 2025

Bengaluru, India

- Architected a full-stack **ML training-data annotation platform** (React, Node.js, MongoDB) used daily by the AI team to label and curate data for production models, replacing spreadsheet workflows with secure role-based REST APIs; cut manual annotation work by **100+ hours/month** and improved labeling throughput by **60%**.
- Engineered a multi-tenant NHS eTriage backend (Node.js, MongoDB) for **8+ UK clients** processing **10,000+ monthly submissions**; owned reliability across releases and remediated **20+ VAPT findings**.

Cario Growth Services Pvt. Ltd.

Backend Developer Intern

November 2022 – May 2023

Chennai, India

- Designed and implemented **20+ RESTful APIs** in Node.js + Fastify backed by PostgreSQL, with input validation and clear contracts used in production by the frontend team.
- Integrated **LLM-powered** context-aware replies into a chat product, iterating on prompt design and response-quality heuristics, and migrated the codebase from JavaScript to TypeScript across multiple repositories.

Projects

Generative Models & Deep Learning - From-Scratch ML

github.com/Deva-1903/UMASS_CICS_689

JAX, PyTorch, NumPy | UMass CS 689 Advanced Machine Learning

- Implemented a **normalizing flow with coupling layers** and a **DDPM diffusion model** from scratch in JAX, deriving the ELBO and closed-form Gaussian-KL objectives via variational inference.
- Trained and benchmarked **5 neural architectures** (perceptron, deep MLP, ReLU MLP, VGG-style CNN, ResNet) on **CIFAR-10** across **3 optimizers** with learning-rate tuning, and built a **reverse-mode autodiff engine** from scratch in NumPy validated against JAX to machine precision.

Spark ETL Performance Optimization - Large-Scale Ingestion

github.com/Deva-1903/Spark-ETL-Optimization

Python, Apache Spark / PySpark, Spark UI, SQL | ~1.4B-row NYC TLC dataset

- Built and optimized a Spark **ETL ingestion pipeline** over a **~1.4B-row** dataset with schema-aware batched reads, column pruning, year/month partitioning with Snappy compression, and a 6-rule **data-quality filter**, cutting end-to-end runtime **~25%** (17–19 min to 14.3 min).
- Replaced shuffle joins with broadcast joins (minutes to ~0.05s), enabled Adaptive Query Execution and skew-join handling, and used **Spark UI** stage timings to localize bottlenecks, cutting tail-latency ratio (P99/P50) from **2.14x to 1.74x**.

AgenticSearch - Provenance-First Entity Discovery

github.com/Deva-1903/ciir_agentic_search

Python, FastAPI, Brave Search, OpenAI / Groq, SQLite

- Built a multi-stage agentic discovery pipeline (typed retrieval planning, page rerank, deterministic extraction with **LLM** fallback, output verification) that returns **cell-level provenance** for every result, deployed as a live demo on **Google Cloud Run**.
- Hardened the service with **150+ automated tests**, provider routing and fallback across **OpenAI and Groq** APIs, and pipeline instrumentation for inspecting extraction quality.

Publication & Extracurriculars

- Publication:** “Alzheimer’s Disease Classification using Transfer Learning,” **IEEE CONIT 2023**, applied deep transfer learning on neuroimaging data for medical image classification.
- Selected for **GirlScript Summer of Code (GSSoC) 2023** and contributed to multiple open-source projects (Linkfree, Freehit, ProjectsHut).
- Tutored **30+** underprivileged students in programming at Sayur, non-profit in South Tamil Nadu.